

Yessin Kcharem

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Les Berges du Lac, Tunis 1053, Tunisia

Summary

Software Engineer with 4 years of experience in backend development, distributed data systems, and real-time services using Python, Scala, Docker, and Kubernetes. Focused on building scalable microservices with strong observability and performance, paired with excellent communication skills and a passion for learning new technologies.

Experience

- ELYADATA** [🌐] Aug 2025 - Present
Tunis, Tunisia
Software Engineer II
 - Developed a real-time voice communication system enabling cellular phone calls between clients and an AI-powered virtual assistant using Twilio, Langflow, and OpenAI realtime agents.
 - Designed a generic, scalable data loader framework capable of ingesting large volumes of structured and semi-structured data from cloud storage and files (e.g. S3, CSV, Parquet), validating records via configurable business rules, and orchestrating distributed processing across multiple Kubernetes pods in parallel.
 - Built end-to-end observability by exposing Prometheus metrics, building Grafana dashboards, and integrating Elastic APM tracing for real-time monitoring, performance analysis, and improved reliability of the virtual assistant.
 - Conducted performance benchmarking with Gatling/Locust to measure the system's ability to handle up to 1000 requests per second under load.
 - Monitored and profiled JVM and PostgreSQL performance with Java Flight Recorder (JFR) and `pg_stat_statements`, tracking slow queries and open database connections to surface performance hotspots.
 - Identified and resolved performance bottlenecks across key system architecture components using observability tools and monitoring insights, resulting in improved scalability and reduced response time by 50%.**Tools:** Python, FastAPI, PostgreSQL, Twilio, Langflow, OpenAI, Prometheus, Grafana, Elastic APM, JFR, `pg_stat_statements`, Gatling, Locust, Docker, Kubernetes, Helm, REST APIs, Microservices
- COGNIRA** [🌐] July 2022 - July 2025
Tunis, Tunisia
Software Engineer II
 - Designed and developed backend features for a complex promotion-planning platform, including a key client feature, "versioning at time of finalization".
 - Architected and deployed caching systems that reduced data retrieval times by 30%, significantly enhancing user experience.
 - Designed and optimized data models for PostgreSQL and Cassandra, ensuring efficient data storage, retrieval, and scalability.
 - Conducted comprehensive unit testing with 80% code coverage, ensuring high code quality and application reliability, identifying and resolving 95% of potential issues prior to production.
 - Migrated backend services build system from SBT to Bazel, improving build performance, dependency management, and CI/CD pipeline efficiency.
 - Developed PySpark migration scripts for efficient database migration and data transformation.**Tools:** Scala, PostgreSQL, Cassandra, Kubernetes, Docker, Microservices, REST APIs, Akka, Git, Bitbucket, Agile development
- TELNET Groupe** [🌐] Jan 2022 - July 2022
Sfax, Tunisia
Software Engineer
 - Led the development of a resume scanning and classification solution using OCR and AI technologies, following Agile Scrum for timely delivery and requirement alignment.
 - Built backend microservices for classification, authentication, and document management using Python, Flask, and OpenCV, securing access with JWT authorization.
 - Optimized microservice performance with multithreading, reducing document classification processing time by 30%.
 - Developed a Flutter mobile app for scanning resumes, secure authentication, and document management, integrating contour detection to improve scan accuracy.**Tools:** Python, Flask, Flutter, NLP, MySQL, JWT, OAuth 2.0, Git, GitLab

Internships

- **Future Proof**

June 2021 - Aug 2021

Intern

Sfax, Tunisia

- Led the study, design, and implementation of a business process for managing a real estate advertisement database during a summer internship at Future Proof.
- Benchmarked various web scraping tools to configure and gather real estate advertisements from multiple websites.
- Utilized Flowable, BPMN, Named Entity Recognition (NER), and Web Scraping to streamline data collection and quality control processes.
- Developed backend microservices with Spring Boot to process and analyze data, integrating machine learning models to extract key features from over 1,000 textual descriptions.
- Designed and implemented a PostgreSQL data model to support seamless storage and retrieval for a database with over 50,000 records.

Tools: Java, Spring Boot, PostgreSQL, Flowable, Microservices, Machine Learning, Agile Development, BPMN 2.0

Skills

- **Programming Languages:** Scala, Python, Java
- **Frameworks & Libraries:** FastAPI, Flask, Spring Boot, Akka, PySpark
- **DevOps & Version Control:** Git, Bitbucket, GitLab, Docker, Kubernetes, Helm, Jenkins, Bazel
- **Databases:** PostgreSQL, Cassandra, MySQL
- **Monitoring & Observability:** Prometheus, Grafana, Elastic APM, JFR, pg_stat_statements
- **Testing & Code Quality:** Gatling, Locust, SonarQube
- **AI & Real-time:** OpenAI, Langflow, Twilio
- **Data Science & Machine Learning:** NLP, Pandas, spaCy
- **Cloud & Mobile:** Azure, Flutter, Android Studio

Education

- **National School of Computer Science (ENSI)**

Sept 2019 - June 2022

Engineering Degree in Computer Science ([Accredited by ASIIN with the EUR-ACE label](#))

Tunis, Tunisia

- Grade: 16.53/20 (Highest Honors)

- **Preparatory Institute for Engineering Studies of Sfax (IPEIS)**

Sept 2017 - June 2019

Pre-Engineering Degree

Sfax, Tunisia

Honors and Awards

- **Promoted to Software Engineer II** – Advanced to the role within one year at COGNIRA, 2023.
- **Technical Interviewer** – Actively participated in conducting technical interviews at COGNIRA, 2023.
- **Top 5% Academic Distinction** – National School of Computer Science (ENSI), 2021.
- **Ranked Top 7% Nationwide** – National Engineering Entrance Exam (Concours Nationaux d'Entrée aux Cycles d'Ingénieurs), Tunisia, 2019.
- **Top Student Award** – Ranked 1st in class during Baccalaureate year, Tunisia, 2017.

Languages

- **English:** Advanced
- **French:** Intermediate
- **German (A1):** Beginner
- **Arabic:** Native